in the Laplace transform by introducing a new complex variable, *s*, defined to be: $s = \sigma + j\omega$. In this same way, we will define a new variable for the z-transform:

$$z = re^{-j\omega}$$

This is defining the complex variable, *z*, as the polar notation combination of the two real variables, *r* and ω. The third step in deriving the z-transform is to replace: *r* and ω, with *z*. This produces the standard form of the z-transform:

EQUATION 33-1
The z-transform. The z-transform defines the relationship between the time domain signal, $x[n]$, and the z-domain signal, $X(z)$.

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Why does the z-transform use $r^n$ instead of $e^{-\sigma n}$, and *z* instead of *s*? As described in Chapter 19, recursive filters are implemented by a set of *recursion coefficients*. To analyze these systems in the z-domain, we must be able to convert these recursion coefficients into the z-domain *transfer function*, and back again. As we will show shortly, defining the z-transform in this manner ($r^n$ and *z*) provides the simplest means of moving between these two important representations. In fact, defining the z-domain in this way makes it *trivial* to move from one representation to the other.

Figure 33-2 illustrates the difference between the Laplace transform's s-plane, and the z-transform's z-plane. Locations in the s-plane are identified by two parameters: σ, the exponential decay variable along the horizontal axis, and ω, the frequency variable along the vertical axis. In other words, these two real parameters are arranged in a *rectangular* coordinate system. This geometry results from defining *s*, the complex variable representing position in the s-plane, by the relation: $s = \sigma + j\omega$.

In comparison, the z-domain uses the variables: *r* and ω, arranged in *polar* coordinates. The distance from the origin, *r*, is the value of the exponential decay. The angular distance measured from the positive horizontal axis, ω, is the frequency. This geometry results from defining *z* by: $z = re^{-j\omega}$. In other words, the complex variable representing position in the z-plane is formed by combining the two real parameters in a polar form.

These differences result in vertical *lines* in the s-plane matching *circles* in the z-plane. For example, the s-plane in Fig. 33-2 shows a pole-zero pattern where all of the poles & zeros lie on vertical lines. The equivalent poles & zeros in the z-plane lie on circles concentric with the origin. This can be understood by examining the relation presented earlier: $\sigma = -\ln(r)$. For instance, the s-plane's vertical axis (i.e., $\sigma = 0$) corresponds to the z-plane's